

I am working with the Titanic dataset, which is publicly available in my GitHub here-
<https://raw.githubusercontent.com/NafeesKhandker/titanic/refs/heads/main/titanic.csv>

I have cleaned the data, and the first five rows of the `gt()` table are shown below:

	Passengerid	Age	Sex	Pclass	Fare	sibsp	Parch	
	Embarked	2	survived					
1	22.00	0	3	7.2500	1	0	2	0
2	38.00	1	1	71.2833	1	0	0	1
3	26.00	1	3	7.9250	0	0	2	1
4	35.00	1	1	53.1000	1	0	2	1
5	35.00	0	3	8.0500	0	0	2	0

Suggest to me some EDA from this dataset, and also include why it is important in 2/3 sentences.

Here's a concise set of **Exploratory Data Analysis (EDA) suggestions** for your Titanic dataset:

- 1. Survival Rate Analysis** – Calculate the overall survival rate and compare survival across groups (e.g., by `Sex`, `Pclass`, `Embarked`).
Importance: This reveals which factors (gender, class, or port of embarkation) might have influenced survival.

- 2. Age Distribution** – Plot histograms or density plots of `Age` for survivors vs. non-survivors.

Importance: Helps identify age groups with higher or lower survival probabilities.

- 3. Fare vs. Survival** – Compare `Fare` distributions for survivors and non-survivors, maybe with boxplots.

Importance: Shows if ticket price (proxy for socioeconomic status) affected survival.

- 4. Family Size Analysis** – Create a new feature `FamilySize = sibsp + parch + 1` and analyze survival rates by family size.

Importance: Examines the impact of traveling alone vs. with family on survival.

- 5. Correlation Heatmap** – Show correlations among numerical variables (`Age`, `Fare`, `sibsp`, `Parch`) and survival.

Importance: Identifies potential predictors for modeling and highlights multicollinearity.

- 6. Embarked & Pclass Interaction** – Analyze survival rate across combinations of `Embarked` and `Pclass`.

Importance: Reveals patterns in survival based on both socio-economic class and boarding location.

If you want, I can **draft a compact R or Python code snippet** to perform all these EDA steps quickly. It'll save a lot of time. Do you want me to do that?



Give me R code for the following EDA. Note that my dataset name is `df_subset`:

Age Distribution – Plot histograms or density plots of `Age`

for survivors vs. non-survivors.

Importance: Helps identify age groups with higher or lower survival probabilities.

Here's a concise R code snippet using `ggplot2` to visualize the **Age distribution by survival** for your `df_subset` dataset:

```
r

# Load required library
library(ggplot2)

# Convert 'Survived' to a factor for easier plotting
df_subset$Survived <- factor(df_subset$Survived, labels = c("No", "Yes"))

# Histogram of Age by Survival
ggplot(df_subset, aes(x = Age, fill = Survived)) +
  geom_histogram(alpha = 0.6, position = "identity", bins = 30) +
  labs(title = "Age Distribution by Survival",
       x = "Age",
       y = "Count",
       fill = "Survived") +
  theme_minimal()

# Density plot of Age by Survival
ggplot(df_subset, aes(x = Age, color = Survived, fill = Survived)) +
  geom_density(alpha = 0.3) +
  labs(title = "Age Density by Survival",
       x = "Age",
```

```
y = "Density",  
fill = "Survived",  
color = "Survived") +  
theme_minimal()
```

✅ Explanation:

- The **histogram** shows the count of passengers in each age group, separated by survival.
- The **density plot** gives a smooth distribution, making it easier to see differences in age patterns between survivors and non-survivors. [Copy code](#)

If you want, I can also **add a combined plot with side-by-side faceting**, which often looks cleaner for presentations. Do you want me to do that?



Do the following EDA next and provide R code:

Survival Rate Analysis – Calculate the overall survival rate and compare survival across groups (e.g., by Sex, Pclass, Embarked).

Here's a clean R snippet to perform **Survival Rate Analysis** for your `df_subset` dataset, including overall survival and breakdowns by `Sex`, `Pclass`, and `Embarked`:

```
r
```

[Copy code](#)

```

# Load required Library
library(dplyr)
library(ggplot2)

# Convert 'Survived' to factor
df_subset$Survived <- factor(df_subset$`Survived`, labels = c("No", "Yes"))

# 1 Overall survival rate
overall_survival <- df_subset %>%
  summarise(Survival_Rate = mean(`Survived`))
print(overall_survival)

```

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```

# 2 Survival rate by Sex
survival_by_sex <- df_subset %>%
  group_by(Sex) %>%
  summarise(Survival_Rate = mean(`Survived`))
print(survival_by_sex)

# 3 Survival rate by Pclass
survival_by_pclass <- df_subset %>%
  group_by(Pclass) %>%
  summarise(Survival_Rate = mean(`Survived`))
print(survival_by_pclass)

# 4 Survival rate by Embarked
survival_by_embarked <- df_subset %>%
  group_by(Embarked) %>%
  summarise(Survival_Rate = mean(`Survived`))
print(survival_by_embarked)

```

```
# Optional: Visualize survival by Sex, Pclass, and Embarked
ggplot(df_subset, aes(x = factor(Sex), fill = Survived)) +
  geom_bar(position = "fill") +
  ylab("Proportion") +
  labs(title = "Survival Rate by Sex", x = "Sex") +
  theme_minimal()

ggplot(df_subset, aes(x = factor(Pclass), fill = Survived)) +
  geom_bar(position = "fill") +
  ylab("Proportion") +
  labs(title = "Survival Rate by Pclass", x = "Passenger Class") +
  theme_minimal()

ggplot(df_subset, aes(x = factor(Embarked), fill = Survived)) +
  geom_bar(position = "fill") +
  ylab("Proportion") +
  labs(title = "Survival Rate by Embarked", x = "Port of Embarkation") +
  theme_minimal()
```

✅ Explanation:

- `mean(survived)` calculates the proportion of survivors (0 = no, 1 = yes).
- `group_by()` splits the dataset into groups (Sex , Pclass , Embarked) to compare survival rates.
- `geom_bar(position = "fill")` visualizes **proportions** instead of counts, making differences between groups clear.

If you want, I can also **combine all three plots into one faceted plot**, which is great for a single slide view. Do you want me to do that?



